# M13 Batch 8 — Full Set

M13.8 Instruments (ATA 31) · PDF pages 651–767

Artificial horizons, attitude director indicators, turn and slip indicators, directional gyros, magnetism and compass systems.

271 questions · Level 3: 57 · Level 2: 142 · Level 1: 72

● marks a question built on a fact your instructor highlighted or underlined in the manual (221 of 271). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

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## Artificial horizons

PDF pages 651–663 · 32 questions

### LEVEL 1 — RECALL

1. Which two tasks are carried out by the gimballed gyros of an aircraft ●
   a. attitude measurement and directional measurement
   b. attitude measurement and vibration measurement
   c. airspeed measurement and directional measurement

2. An attitude gyro is running normally in the cruise. Changes in the angle of which two aircraft axes does it measure? ●
   a. the longitudinal axis and the vertical axis
   b. the lateral axis and the vertical axis
   c. the longitudinal axis and the lateral axis

3. What is the result of the measurement made by the attitude gyro ●
   a. the angle of yaw and drift
   b. the angle of roll and pitch
   c. the angle of flight direction

4. Which gyro gives the reference for the attitude indication of an aircraft ●
   a. a rate gyro
   b. a vertical gyro
   c. a directional gyro

5. Which other name is given to the attitude indication of an aircraft ●
   a. the attitude director
   b. the horizon drum
   c. the artificial horizon

6. A standby horizon is being powered up for a check. From which aircraft supply does its gyro take its drive energy? ●
   a. the battery system
   b. the main AC generator
   c. the cockpit suction pump

7. How does the erection system of the standby horizon do its work ●
   a. pneumatically
   b. electrically
   c. mechanically

8. The erection system of a standby horizon can influence the gyro axis to a maximum rate of ●
   a. ten degrees per minute
   b. three degrees per minute
   c. thirty degrees per minute

9. A technician pulls the caging knob of a standby horizon. What happens inside the instrument? ●
   a. the gyro motor is switched off and allowed to run down
   b. the erection balls are released onto their circular path
   c. the gimbals are caged and stabilized in the normal position

10. What kind of motor is the gyro of a ball driven standby horizon ●
    a. a single-phase synchronic external rotor motor
    b. a three-phase asynchronic internal rotor motor
    c. a three-phase asynchronic external rotor motor

11. The gyro motor of a ball driven standby horizon loses its electrical supply. What shows this on the instrument? ●
    a. a level switch
    b. an ammeter
    c. a flag

12. The gyro of a ball driven standby horizon turns at about ●
    a. 11,000 rpm
    b. 5,000 rpm
    c. 22,000 rpm

## LEVEL 2 — UNDERSTAND AND EXPLAIN

13. A unit on the bench is labelled as a directional gyro. What does that tell a technician about the way its spin axis lies? ●
    a. it is vertical
    b. it is at 45 degrees
    c. it is horizontal

14. A gyro on the bench has its rotational axis standing upright. Which measuring task is it built for? ●
    a. attitude measurement
    b. directional measurement
    c. turn rate measurement

15. Where must the rotational axis of the attitude gyro always point ●
    a. along the flight path
    b. towards the magnetic north pole
    c. towards the centre of the earth

16. When direction is measured, changes in the angle of which aircraft axis are detected ●
    a. the lateral axis
    b. the longitudinal axis
    c. the vertical axis

17. A directional gyro is running correctly in level flight. How does its rotational axis lie? ●
    a. along the lateral axis of the aircraft
    b. straight down at the centre of the earth
    c. parallel to the horizon of the earth

**18.** In a small aircraft, which instrument has the vertical gyro built into it

- **a.** the suction indicator
- **b.** the attitude indicator
- **c.** the standby horizon

**19.** Why must the artificial horizon stay available for as long as possible ●

- **a.** it is the only indication that works without any electrical power
- **b.** it is one of the four most important indications in an aircraft
- **c.** it is the only indication that shows both roll and pitch together

**20.** A standby horizon gyro has run down past the point where it still turns stable. What has happened to its indication? ●

- **a.** it is no longer usable
- **b.** it is corrected by the caging knob
- **c.** it is frozen at the last attitude

**21.** Why does a standby horizon need a caging mechanism as well as its erection system ●

- **a.** erection on its own cannot hold the gimbals in the normal position
- **b.** erection on its own is far too slow to set the gyro up at once
- **c.** erection on its own cannot work while the gyro runs at full speed

**22.** After which event is an immediate erection of the gyro wanted ●

- **a.** power activation
- **b.** a long descent
- **c.** a heavy landing

**23.** At which gyro speeds is gimbal caging allowed to be used ●

- **a.** at any speed between standstill and full speed
- **b.** only while the gyro is running down from full speed
- **c.** at full speed or at a complete standstill

**24.** Apart from the horizon drum, what else is turned by the gyro motor of a ball driven standby horizon ●

- **a.** the erection and erection lock mechanism
- **b.** the pendulum stop and the frame stops
- **c.** the caging knob and the aeroplane symbol

**25.** What is prevented by the cut-out mechanism of a ball driven standby horizon ●

- **a.** the gyroscope running down after a power failure
- **b.** the gyroscope tipping towards apparent vertical
- **c.** the gyroscope precessing towards earth vertical

**26.** An aircraft is loaded heavily so it sits nose high on the ground. How can neutral pitch attitude still be set on the standby horizon? ●

- **a.** by pulling the caging knob to cage the gimbals
- **b.** by turning the caging knob to move the aeroplane symbol
- **c.** by resetting the barometric knob of the instrument

### LEVEL 3 — APPLY AND REASON

**27.** A technician cages the gimbals while the gyro is still coasting down at half speed. What is the likely result? ●

- **a.** the gyro will erect at twice its normal rate
- **b.** the gyro will stop at once and need a restart
- **c.** the gyro may be damaged, stability being limited

**28.** The gyro axis points at the centre of the earth during unaccelerated flight. What are the two erection balls doing? ●

- **a.** they are held at their stops by the gyro cut-out rocker
- **b.** they are driven apart, so a momentum acts on the gyro axis
- **c.** they are balanced out, so no momentum acts on the gyro axis

**29.** A horizontal acceleration greater than 1.7 m/s2 acts on a ball driven standby horizon. What happens inside it? ●

- **a.** a rocker tips and prevents an unbalance of the balls
- **b.** a flag drops to show the indication is no longer usable
- **c.** a torque motor drives the balls apart to correct the axis

**30.** At start-up the gyro axis of a ball driven horizon is found more than 10 degrees out of earth vertical. What must be done? ●

- **a.** pull the caging knob to erect the gyro rapidly
- **b.** wait for the erection balls to bring the axis back
- **c.** switch the supply off and let the gyro run down

**31.** Power to a ball driven standby horizon is lost in flight. For how long can the instrument still be used? ●

- **a.** up to 5 minutes
- **b.** up to 15 minutes
- **c.** up to 45 minutes

**32.** A standby horizon has lost all power in flight. What sets the limit on how long the crew may keep using it? ●

- **a.** the rotational energy still stored in the gyro
- **b.** the charge still left in the aircraft battery
- **c.** the air still being drawn through the instrument

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## Answers — Artificial horizons

**1.** **a** The two important functions named for gimballed gyros are attitude measurement and directional measurement. (L1, PDF p. 651)

**2.** **c** The attitude gyro measures the angle changes of the x-axis and the y-axis. (L1, PDF p. 653)

**3.** **b** Measuring the x and y axis angle changes gives the angle of roll and pitch. (L1, PDF p. 653)

**4.** **b** The vertical gyro is the reference for the attitude indication. (L1, PDF p. 657)

**5.** **c** The attitude indication is also called the artificial horizon. (L1, PDF p. 657)

**6.** **a** The standby horizon gyro is electrically driven from the aircraft battery system. (L1, PDF p. 659)

**7.** **c** The erection system of the standby horizon works mechanically. (L1, PDF p. 659)

**8.** **b** An erection system can only influence the gyro axis to a maximum of three degrees per minute. (L1, PDF p. 661)

**9.** **c** Pulling the caging knob cages the gimbals and stabilizes them in the normal position. (L1, PDF p. 661)

**10.** **c** The gyro is a three-phase asynchronic external rotor motor. (L1, PDF p. 663)

**11.** **c** A flag monitors the electrical supply to the motor. (L1, PDF p. 663)

12. **c** The gyro of the ball driven standby horizon turns at about 22,000 rpm. (L1, PDF p. 663)
13. **c** Gyros with a horizontal axis are the ones called directional gyros. (L2, PDF p. 651)
14. **a** The vertical axis gyro is the attitude gyro, so it does attitude measurement. (L2, PDF p. 651)
15. **c** The attitude gyro axis must always point at the centre of the earth, that is, be vertical. (L2, PDF p. 653)
16. **c** Direction measurement comes from the angle changes of the vertical axis. (L2, PDF p. 653)
17. **c** The directional gyro axis must always be parallel to the earth's horizon, that is horizontal. (L2, PDF p. 653)
18. **b** In small aircraft the vertical gyro sits inside the attitude indicator. (L2, PDF p. 657)
19. **b** It is named as one of the four most important indications, so it must keep working as long as it can. (L2, PDF p. 659)
20. **a** The indication stays usable only as long as the gyro runs stable during its spin down time. (L2, PDF p. 659)
21. **b** Erection moves the axis only three degrees a minute, so a caging mechanism is needed to erect the gyro immediately. (L2, PDF p. 661)
22. **a** Immediate erection is wanted for example after power activation. (L2, PDF p. 661)
23. **c** Caging may be used only with the gyro at full speed or stopped completely. (L2, PDF p. 661)
24. **a** The same motor also drives the erection and the erection lock mechanism. (L2, PDF p. 663)
25. **b** The cut-out mechanism stops the gyroscope from tipping towards apparent vertical. (L2, PDF p. 663)
26. **b** Turning the caging knob raises or lowers the symbol, so neutral pitch can be set whatever the weight. (L2, PDF p. 663)
27. **c** Any intermediate speed may damage the gyro because of the limited stability. (L3, PDF p. 661)
28. **c** With the axis in earth vertical the balls are balanced and no momentum acts on the gyro axis. (L3, PDF p. 663)
29. **a** Above 1.7 m/s2 a rocker tips and keeps the balls from going out of balance. (L3, PDF p. 663)
30. **a** Beyond 10 degrees the cut-out mechanism blocks independent erection, so the caging knob must be pulled. (L3, PDF p. 663)
31. **b** The gyrating mass lets the standby horizon be used for up to 15 minutes after a power failure. (L3, PDF p. 663)
32. **a** The gyrating mass carries the gyro on for up to 15 minutes, and that stored energy is what sets the limit. (L3, PDF p. 663)

# Vertical gyros and erection systems

PDF pages 665–679 · 56 questions

## LEVEL 1 — RECALL

**33.** How far can the mass on one end of the balance beam swing out when the gyro is unbalanced ●

a. up to 85 degrees
b. up to 30 degrees
c. up to 10 degrees

**34.** In a small aircraft, what generates the airflow that drives the gyro ●

a. a vacuum or suction pump
b. an engine bleed tapping
c. a static inverter

**35.** What do the letters ISIS stand for ●

a. Integrated Second Instrument Set
b. Integrated Standby Instrument System
c. Internal Standby Indicating System

**36.** Which three indications are supplied by a recent integrated standby instrument system ●

a. altitude, airspeed and vertical speed
b. attitude, altitude and airspeed
c. attitude, altitude and radio height

**37.** How many degrees of freedom does a vertical gyro have ●

a. one
b. three
c. two

**38.** How are vertical gyros driven ●

a. by an electric motor
b. by cockpit airflow
c. by engine bleed air

**39.** Which switches do electrical gyros carry to deal with aircraft accelerations ●

a. conductive level switches
b. erection cutout switches
c. gimbal caging switches

**40.** Which aircraft axis does the roll axis of the vertical gyro correspond to ●

a. the lateral axis
b. the longitudinal axis
c. the vertical axis

**41.** Which aircraft axis is represented by the pitch axis of the vertical gyro ●

a. the lateral axis
b. the longitudinal axis
c. the vertical axis

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**42.** How does an improved standby horizon obtain the alternating current it needs to run

- **a.** from a static inverter installed inside the device
- **b.** from the standby busbar of the aircraft supply
- **c.** from a phase converter mounted in the aircraft rack

**43.** Which mechanism is absent from a pendulum erection standby horizon ●

- **a.** the ball circulating mechanism
- **b.** the gimbal caging mechanism
- **c.** the mechanical cut-out mechanism

**44.** What takes the place of the ball circulating mechanism in a pendulum erection horizon ●

- **a.** a pendulum accelerometer and rocker
- **b.** a balance beam turning at low speed
- **c.** a pair of conductive level switches

**45.** The gyro of a standby horizon with a balance beam is no longer in its horizontal position and the beam mass swings out. What is that displacement used for? ●

- **a.** as the signal that drops the warning flag
- **b.** as the erection momentum for the gyroscope
- **c.** as the braking force on the gyro capsule

**46.** Besides a large gyro unbalance, what else makes the cut-out mechanism hold the rotational mass in place ●

- **a.** a rise of the gyro rotational speed
- **b.** equivalent horizontal accelerations
- **c.** a drop of the supply voltage

**47.** Why does a gyro need a high rotational speed

- **a.** to shorten its erection time
- **b.** to lower its power consumption
- **c.** to have enough stability

**48.** The suction pump of a small aircraft is running. Which air is being drawn through the indicators? ●

- **a.** the bleed air, through a cooling unit
- **b.** the ram air, through an intake scoop
- **c.** the cockpit air, through a filter

**49.** What shape is given to the gyro body of an air driven instrument ●

- **a.** it is formed like a hollow sphere
- **b.** it is formed like a small turbine
- **c.** it is formed like a flat disc

**50.** Which two advantages does an electric motor drive give to the gyros of commercial aircraft ●

- **a.** a simpler filter fit and a lower running temperature
- **b.** more reliability and a stable speed in all flight situations
- **c.** lower weight and a shorter erection time after switch on

**51.** In what kind of construction are the latest standby horizons built

- **a.** an air driven standby gyro with pendulum erection
- **b.** a mechanically caged standby gyro with a horizon drum
- **c.** a solid state electronic standby instrument system

**52.** Besides attitude, altitude and airspeed, which two further values can a recent integrated standby system display ●

- **a.** magnetic heading and mach number
- **b.** drift angle and ground speed
- **c.** outside air temperature and wind

**53.** An aircraft is fitted with an integrated standby instrument system. What has become of the separate standby instruments? ●

- **a.** they are no longer needed
- **b.** they are moved to the overhead panel
- **c.** they are kept as a second back up

**54.** In which case is a vertical gyro used for an artificial horizon ●

- **a.** when the gyro is not linked mechanically to the dial
- **b.** when the gyro must be driven by cockpit airflow
- **c.** when the gyro has no erection system of its own

**55.** A vertical gyro is mounted in the avionic compartment as an attitude sensor. What else must the installation then have? ●

- **a.** a suction pump in the cockpit
- **b.** an indicator in the cockpit
- **c.** a caging knob in the compartment

**56.** What holds the axis of a vertical gyro in the vertical position ●

- **a.** a caging mechanism
- **b.** a pendulum stop
- **c.** an erection system

**57.** The aircraft rolls while a remote vertical gyro is working. By what is the attitude angle sensed? ●

- **a.** by level switches
- **b.** by torque motors
- **c.** by synchros

**58.** What does an air driven gyro use to generate its erection force ●

- **a.** a balance beam turning slowly
- **b.** level switches holding a fluid
- **c.** pendulums moved by the airflow

**59.** What activates the torque motor of an electrically driven gyro when its axis topples ●

- **a.** a level switch holding a conductive fluid
- **b.** a pendulum moved by the cockpit airflow
- **c.** a rocker tipped by horizontal acceleration

**60.** A force has to be applied to move a gyro axis in a chosen direction. At what angle from that direction must the force act? ●

- **a.** 45 degrees
- **b.** 180 degrees
- **c.** 90 degrees

**61.** Which error must the normal erection rate compensate, besides precessions from real topple ●

- **a.** an apparent topple of 15 degrees per hour
- **b.** an erection drift of 3 degrees per minute
- **c.** a real topple of 20 degrees per minute

**62.** An electrical gyro is switched on again after a long power interruption. Which erection rate is used? ●

a. about 20 degrees per minute
b. about 3 degrees per minute
c. about 2 degrees per minute

**63.** What can a vertical gyro no longer detect once longitudinal accelerations or centrifugal forces act on it ●

a. its own rotational speed
b. the centre of the earth
c. the lateral axis of the aircraft

**64.** During which manoeuvres must the erection system be switched off ●

a. during curves or longitudinal accelerations
b. during climbs and descents at steady speed
c. during taxiing and engine ground running

**65.** An erection cutout switch has sensed a roll axis input while the aircraft manoeuvres. For what period does it keep the erection circuit interrupted? ●

a. until the gyro is caged by the technician
b. for as long as the acceleration lasts
c. for a fixed period of three seconds

**66.** Why must long acceleration periods such as full turns be avoided ●

a. the gyro topple cannot be compensated during that time
b. the erection rate rises above its normal working limit
c. the gyro motor runs hot in that attitude of the frame

**67.** When the attitude of the aircraft changes, how do the aircraft and the gyroscope move in relation to one another ●

a. the gyroscope turns around the aircraft
b. the gyroscope precesses together with the aircraft
c. the aircraft turns around the gyroscope

**68.** How do the gyro axis and the frame axis lie to one another in horizontal flight ●

a. at right angles to one another
b. at 45 degrees to one another
c. parallel to one another

**69.** A vertical gyro is feeding the cockpit indicator. What turns the angular movements at the pivot points into electric signals? ●

a. conductive level switches
b. erection torque motors
c. synchro-transmitters

**70.** What are the voltages between the tapped synchro wires of a locked synchro when the aircraft is in horizontal attitude

a. at half of full value
b. at zero
c. at a maximum

**71.** Which two users take the synchro signals of the vertical gyro ●

a. the attitude director indicator and the autopilot
b. the flight director and the autopilot
c. the weather radar and the flight director

**72.** Which two users are switched to the locked synchros of the vertical gyro ●

a. the weather radar and the flight director
b. the autopilot and the weather radar
c. the attitude director indicator and the autopilot

**73.** By which two means can the alignment of the gyro axis be measured

a. a level switch or a torquer
b. a synchro or a pendulum
c. a plumb line or a spirit level

### LEVEL 3 — APPLY AND REASON

**74.** A pendulum erection horizon is sitting with a gyro unbalance of 12 degrees. What does its cut-out mechanism do? ●

a. it lets the rotational mass swing out to full travel
b. it breaks the electrical supply to the gyro motor
c. it holds the rotational mass in its normal position

**75.** The suction regulator of an air driven gyro system fails and the suction rises well above its normal value. What is the danger to the gyro?

a. it will run down earlier than usual
b. it will run into an overspeed
c. it will topple to apparent vertical

**76.** A gyro instrument is found with a filter and a suction line behind it. In which class of aircraft is it fitted? ●

a. a commercial aircraft
b. a small aircraft
c. an aircraft with no battery

**77.** A vertical gyro sits in the avionic compartment and its synchros are working normally. What makes use of the signal they send? ●

a. the erection cutout switches
b. the indicator or an indication system
c. the supply converter of the gyro motor

**78.** The roll level switch of an electrically driven gyro detects the axis in an incorrect direction. What follows? ●

a. the roll cutout switch opens and the erection circuit is broken
b. the roll torque motor applies a force and precession corrects it
c. the caging mechanism locks the gimbals in a normal position

**79.** An aircraft makes normal turns and accelerations in the cruise. Why does the erection system build up no significant error? ●

a. because the rate it works at is set by hand
b. because the rate it works at is doubled
c. because the rate it works at is so small

**80.** Aircraft in a hold fly racetrack patterns rather than continuous turns. What does the shape of that pattern give the erection system? ●

a. a continuous cutout of both of the erection circuits
b. a steady centrifugal force to erect the gyro against
c. time after each half turn to compensate gyro topple

**81.** The attitude of the aircraft changes while a vertical gyro is working. Which part of the synchro moves with the airframe? ●

- **a.** the rotor, around the stator held steady by the gyro
- **b.** the stator, around the rotor held steady by the gyro
- **c.** both stator and rotor, around the fixed gyro case

**82.** A deviation voltage from a locked synchro changes its phase and grows in amplitude. What do those two changes each tell you? ●

- **a.** the phase gives the axis and the amplitude the rate
- **b.** the phase gives the direction and the amplitude the size
- **c.** the phase gives the size and the amplitude the direction

**83.** A vertical gyro is sitting on the shelf with no power on it. Which stop keeps the gyroscope out of an extreme inclination?

- **a.** the pendulum stop
- **b.** the pitch stop
- **c.** the frame stop

**84.** A vertical gyro has been taken to about 85 degrees of movement around the lateral axis of the aircraft. What holds it there?

- **a.** the pitch stop
- **b.** the pendulum stop
- **c.** the frame stop

**85.** The two variable windings of the torquer are carrying equal currents. Why is the gyro axis left untouched?

- **a.** the magnetic fields are equal and opposite, so no torque results
- **b.** the magnetic fields are unequal, so the two torques cancel out
- **c.** the magnetic fields are equal and additive, so no torque results

**86.** An erection circuit stays active while the aircraft accelerates. Where is the gyro led, and what does the crew see?

- **a.** towards true perpendicular, and a flag in place of the display
- **b.** towards the pendulum stop, and a frozen reading on the display
- **c.** towards apparent vertical, and an equally large error on the display

**87.** A gyro responds 90 degrees away from an applied force. How is that allowed for in the way the torquers are connected?

- **a.** the pitch torquer puts its torque on the yaw axis
- **b.** the pitch torquer puts its torque on the roll axis
- **c.** the pitch torquer puts its torque on the pitch axis

**88.** An aircraft is standing still on the ramp with its vertical gyro powered. Why may the erection circuit work freely here?

- **a.** only the cut out switches are disconnected here
- **b.** only the gyro is turning below its normal speed
- **c.** only the weight is acting as an acceleration

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## Answers — Vertical gyros and erection systems

**33.** **b** The mass on one end can swing out up to 30 degrees in the case of gyro unbalance. (L1, PDF p. 665)

**34.** **a** The airflow comes from a vacuum or suction pump. (L1, PDF p. 667)

**35.** **b** ISIS is the Integrated Standby Instrument System. (L1, PDF p. 669)

36. **b** Recent systems supply attitude, altitude and airspeed indications. (L1, PDF p. 669)
37. **b** A vertical gyro is a gyro with three degrees of freedom. (L1, PDF p. 671)
38. **a** Vertical gyros are driven electrically. (L1, PDF p. 671)
39. **b** Electrical gyros have erection cutout switches for this. (L1, PDF p. 675)
40. **b** The roll axis of the gyro corresponds to the aircraft's longitudinal axis. (L1, PDF p. 677)
41. **a** The pitch axis of the gyro represents the lateral axis of the aircraft. (L1, PDF p. 677)
42. **a** A static inverter built into the device supplies the AC needed for operation. (L2, PDF p. 665)
43. **a** These devices do not have a ball circulating mechanism. (L2, PDF p. 665)
44. **b** In its place a balance beam rotates at low speed. (L2, PDF p. 665)
45. **b** The mass displacement is used as erection momentum for the gyroscope. (L2, PDF p. 665)
46. **b** Equivalent horizontal accelerations have the same effect as a 10 degree unbalance. (L2, PDF p. 665)
47. **c** High rotational speed is what gives the gyro enough stability. (L2, PDF p. 667)
48. **c** The pump sucks cockpit air via a filter through the indicators. (L2, PDF p. 667)
49. **b** The gyro body is formed like a small turbine so the airflow can drive it. (L2, PDF p. 667)
50. **b** Electric drive is stated to be more reliable and to give a stable speed in all flight situations. (L2, PDF p. 667)
51. **c** The latest standby horizons are integrated into a solid state electronic standby instrument system. (L2, PDF p. 669)
52. **a** Magnetic heading and mach number can be displayed in addition. (L2, PDF p. 669)
53. **a** With the integrated system the individual original standby instruments are no longer needed. (L2, PDF p. 669)
54. **a** The vertical gyro is used where the gyro is not mechanically linked to an instrument's indication. (L2, PDF p. 671)
55. **b** With the gyro in the avionic compartment an indicator is needed in the cockpit. (L2, PDF p. 671)
56. **c** An erection system holds the gyro axis vertical. (L2, PDF p. 671)
57. **c** The angle of attitude is sensed by synchros. (L2, PDF p. 671)
58. **c** The air driven gyro uses pendulums to make an erection force from the airflow. (L2, PDF p. 673)
59. **a** Electrically driven gyros use level switches with a conductive fluid to activate the torque motor. (L2, PDF p. 673)
60. **c** Precession means forces must be applied 90 degrees from the result to be achieved. (L2, PDF p. 673)
61. **a** The rate is sufficient to compensate the apparent topple of 15 degrees per hour plus precessions from real topple. (L2, PDF p. 673)
62. **a** A faster erection rate of about 20 degrees per minute is activated after a long power interruption. (L2, PDF p. 673)
63. **b** Under those forces the gyro can no longer detect the centre of the earth. (L2, PDF p. 675)
64. **a** The erection system must be switched off during curves or longitudinal accelerations. (L2, PDF p. 675)
65. **b** The switch interrupts the corresponding erection circuit for as long as the accelerations last. (L2, PDF p. 675)
66. **a** During a long acceleration the erection is cut out, so gyro topple cannot be compensated. (L2, PDF p. 675)
67. **c** Because the gyro is space stable, any change in attitude is a rotating of the aircraft around the gyroscope. (L2, PDF p. 677)
68. **a** In horizontal flight the gyro axis and the frame axis are at right angles. (L2, PDF p. 677)
69. **c** Synchro-transmitters at the pivot points make the angular movements into electric signals. (L2, PDF p. 677)
70. **b** Those voltages are zero at horizontal attitude of the aircraft. (L2, PDF p. 677)
71. **a** The ADI and the autopilot use the synchro signals. (L2, PDF p. 677)
72. **a** WX and FD are switched to the locked synchros. (L2, PDF p. 677)
73. **c** The alignment can be measured with a plumb line or a spirit level. (L2, PDF p. 679)
74. **c** At an unbalance of 10 degrees or more the cut-out mechanism retains the rotational mass in its normal position. (L3, PDF p. 665)
75. **b** The regulator is there to prevent gyro overspeed, so losing its control lets the gyro overspeed. (L3, PDF p. 667)
76. **b** Air driven gyros with a suction pump and filter belong to small aircraft, while commercial aircraft use electric drive. (L3, PDF p. 667)
77. **b** The synchro signal goes to the cockpit indicator or to an indication system. (L3, PDF p. 671)

78. **b** The roll level switch activates the roll torque motor, whose force precesses the gyro back into the correct direction. (L3, PDF p. 673)
79. **c** The erection rate is so small that normal turns and accelerations cannot build a significant error. (L3, PDF p. 673)
80. **c** The racetrack gives the erection system enough time after each half turn to compensate any gyro topple. (L3, PDF p. 675)
81. **b** The synchro-stator turns with the aircraft around the synchro-rotor, which the gyroscope keeps stabilised. (L3, PDF p. 677)
82. **b** Phase shift represents the direction of the deviation and amplitude its dimension. (L3, PDF p. 677)
83. **a** The pendulum stop is the one that prevents an extreme inclination when the gyro is not energised. (L3, PDF p. 677)
84. **a** The pitch stop limits movement around the lateral axis to about +/- 85 degrees. (L3, PDF p. 677)
85. **a** With the same currents the magnetic fields are equal but opposite, and the torquer produces no torque. (L3, PDF p. 679)
86. **c** With erection active during acceleration the gyro is led to apparent vertical, and the axis deviation shows as an equally large error on the horizon display. (L3, PDF p. 679)
87. **b** To use precession, the pitch torquer is made to act on the roll axis and the roll torquer on the pitch axis. (L3, PDF p. 679)
88. **c** In a stationary condition only the earth's acceleration has an effect, which is the condition the erection circuit needs. (L3, PDF p. 679)

# Attitude director indicators

PDF pages 681–693 · 43 questions

## LEVEL 1 — RECALL

**89.** Which supply voltage is fed to the vertical gyro ●

a. 115 V AC
b. 28 V DC
c. 90 V AC

**90.** How long does the fast erection phase last after the vertical gyro is switched on ●

a. about 30 seconds
b. about 2 - 3 minutes
c. about 10 minutes

**91.** Where is the vertical gyro that supplies the attitude signals to the ADI installed ●

a. in the cockpit overhead panel
b. inside the ADI housing itself
c. in the electric compartment

**92.** How many independent vertical gyros are needed so that each pilot has an attitude indication of his own ●

a. one
b. three
c. two

**93.** Which other aircraft systems are supplied with attitude information from the vertical gyros ●

a. the air data system and the instrument warning system
b. the autoflight system and the weather radar system
c. the instrument landing system and the radio altimeter

**94.** Which two kinds of information does the attitude director indicator mainly show ●

a. heading and indications from the flight director system
b. attitude and indications from the flight director system
c. attitude and indications from the instrument landing system

**95.** To which group of main flight instruments does the attitude director indicator belong ●

a. the pitot-static group
b. the Basic T
c. the engine indicating group

**96.** What kind of instrument is the slip indicator fitted to the ADI ●

a. a mechanical ball in a liquid filled curved glass tube
b. an electric servo motor fed from the vertical gyro
c. a pendulum vane damped by a light return spring

**97.** Which rate of turn does the turn indication on the ADI represent ●

a. 360°/min
b. 90°/min
c. 180°/min

**98.** Why is the attitude comparison fitted to the attitude reference system ●

- **a.** as a means of erecting the gyros faster
- **b.** as a replacement for the third gyro
- **c.** as an additional safety feature

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**99.** Which condition must have applied before switch on for the vertical gyro to run a fast erection phase ●

- **a.** the erectors had been held at the reduced support rate
- **b.** the gyro had been without current for more than 30 seconds
- **c.** the gyro motor had been running above 75% rpm

**100.** A vertical gyro has held a motor speed above 75% for more than 30 seconds and its gimbals are erected. What happens next? ●

- **a.** fast erection ends and the motor runs on at reduced voltage
- **b.** the erectors are switched off for a maximum of three minutes
- **c.** fast erection begins and the motor runs on at raised voltage

**101.** In the vertical gyro, strong horizontal acceleration switches off one erector and strong longitudinal acceleration the other. Which way round is it? ●

- **a.** horizontal switches off the roll erector, longitudinal the pitch erector
- **b.** both of the two accelerations switch off the pitch erector
- **c.** horizontal switches off the pitch erector, longitudinal the roll erector

**102.** What is the longest time the pitch erector may stay switched off during strong longitudinal acceleration ●

- **a.** 30 seconds
- **b.** three minutes
- **c.** two minutes

**103.** A flag monitor is checking the gyro supply before it releases the valid voltage. What level must that supply exceed?

- **a.** 115 V
- **b.** 75 V
- **c.** 90 V

**104.** A technician traces the ADI pitch and roll signals back to the vertical gyro gimbals. What will he find converting the angles into electrical signals? ●

- **a.** servo motors
- **b.** synchros
- **c.** moving coils

**105.** An ADI receives a gyro valid signal from its vertical gyro. What has the gyro confirmed by sending it?

- **a.** that its gimbals are free of friction
- **b.** that it is running at its normal speed
- **c.** that its synchros are correctly aligned

**106.** One of the two normal vertical gyros has failed in flight. How is that pilot's attitude indication restored? ●

- **a.** his indicator falls back on the standby horizon signal
- **b.** a switching transfers the third vertical gyro to his indicator
- **c.** his indicator is cross-wired to the other pilot's vertical gyro

**107.** A gyro has failed and the spare is to be brought in. To which indicators can that third gyro be switched? ●

a. to either of the two pilots' ADIs
b. to both ADIs at the same time
c. to one fixed ADI only

**108.** The aeroplane symbol of the ADI follows every movement of the aircraft. Why does it do that? ●

a. it is driven by the roll synchro of the vertical gyro
b. it is carried on the instrument housing itself
c. it is mounted on the fork that holds the horizon ball

**109.** How does the blue and black horizon ball of the ADI behave as the aircraft manoeuvres

a. it turns with the aircraft movement and is kept horizontal by the gyro
b. it turns contrary to the aircraft movement and is kept horizontal by the gyro
c. it is held still while the pitch and roll scales move around it

**110.** What does the VG valid direct current arriving at the ADI confirm ●

a. that the vertical gyro is functioning
b. that the synchro signals have nulled
c. that the ADI power supply is healthy

**111.** What makes the ball of the slip indicator take up the apparent vertical ●

a. the yaw rate signal fed to a coil behind the ball
b. the influence of gravity on the ball
c. the deviation signal of the navigation receiver

**112.** A technician watches the ADI turn indication while the aircraft manoeuvres. Which change in the aircraft's movement does that indication respond to? ●

a. a change in rotational velocity about the vertical axis
b. a change in the resultant force acting on the aircraft
c. a change in rotational velocity about the lateral axis

**113.** Which kind of instrument movement drives the turn indication on the ADI ●

a. a moving coil
b. a synchro receiver
c. an electric servo motor

**114.** How much deviation from the localizer do the marks beside the runway symbol stand for

a. about three degrees left or right
b. about one degree left or right
c. about one third of a degree left or right

**115.** What do the flight director bars show the pilot ●

a. the direction in which the aircraft is to be flown
b. the deviation of the aircraft from the runway centreline
c. the direction in which the aircraft is actually moving

**116.** From where does the attitude comparison take the signals it works on

a. directly from the autoflight system
b. directly from the indication
c. directly from the vertical gyro gimbals

**117.** How far apart must the pitch or roll indications be before the alert is raised ●

a. several minutes of arc apart
b. any amount apart at all
c. several degrees apart

**118.** Which system raises the alert when the two attitude indications disagree ●

a. the weather radar system
b. the instrument warning system
c. the autoflight system

**119.** An aircraft has primary flight displays in place of ADIs and the two attitudes disagree. What do the crew see? ●

a. a CHECK ATTITUDE message above the attitude indication on one display
b. a CHECK ATTITUDE message above the attitude indication on both displays
c. an attitude flag across the attitude indication on both displays

**120.** Which sources can serve as the third attitude reference for fault isolation ●

a. vertical gyro number 3 or the standby horizon
b. vertical gyro number 1 or the standby horizon
c. vertical gyro number 3 or the weather radar

## LEVEL 3 — APPLY AND REASON

**121.** A vertical gyro has just reported valid and the voltage on its erectors has dropped. What support rate are the erectors working at now? ●

a. 2°/min
b. 20°/min
c. 28°/min

**122.** A vertical gyro is at full speed with its supply voltage in limits and its output synchros energised, yet the flag monitor still withholds the valid voltage. Which remaining condition is missing?

a. the roll erector has not yet been switched off
b. fast erection has not yet been completed
c. the supply has not yet reached its rated alternating voltage

**123.** An aircraft carries a second autopilot as well as its original one. Which vertical gyro feeds that second system? ●

a. the first officer's vertical gyro
b. the third vertical gyro
c. the captain's vertical gyro

**124.** The nose of the aircraft is raised. How does the horizon ball of the ADI respond to that movement?

a. it is swung the opposite way round by the fork
b. it is held still while the symbol climbs
c. it spins about its own axis inside the fork

**125.** A pilot flies the flight director bars while watching the horizon ball on the same indicator. What do the two indications give him together? ●

a. where the aircraft is now and how fast it is turning
b. how the aircraft is banked and where the runway lies
c. where the aircraft is now and where it is to be flown

**126.** On an ILS approach the runway symbol is seen rising towards the aeroplane symbol. What is it telling the crew?

- **a.** the radio altitude is decreasing
- **b.** the localizer deviation is increasing
- **c.** the glideslope is being captured

**127.** An ADI is being used as the aircraft's rate of turn indicator. Which movement feeds that indication and what always accompanies it? ●

- **a.** the yaw rate, accompanied by the flight director bars
- **b.** the yaw rate, accompanied by the ball
- **c.** the roll rate, accompanied by the ball

**128.** A technician presses the ATT TEST button and the ATT flag comes into view. What does that tell him?

- **a.** the test voltage never reached the indicator synchros
- **b.** the null monitoring brings the flag in as part of the test
- **c.** the indicator is faulty, as the flag should stay out of view

**129.** An ADI horizon ball has swung round to a 90° bank attitude. Which condition can produce this on some horizon indicators?

- **a.** the power supply has failed or the valid signal is missing
- **b.** the attitude test button has been pressed and held in
- **c.** the expanded part of the pitch scale has been run off

**130.** An ADI's two synchro systems null correctly and its gyro reports valid, but the indicator has lost its power supply. What does that mean for the attitude display?

- **a.** the synchros still hold the ball in its last position
- **b.** the valid signal on its own keeps the display alive
- **c.** it cannot be shown, as all the conditions must be met

**131.** Two attitude indications disagree and the crew must work out which of them to believe. What does that always need? ●

- **a.** a third independent source of the same information
- **b.** the valid signal taken from each of the two gyros
- **c.** a repeat of the attitude comparison against the first source

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## Answers — Attitude director indicators

**89.** **a** The voltage supplied to the vertical gyro is 115 V AC. (L1, PDF p. 681)

**90.** **b** The fast erection phase lasts about 2 - 3 minutes. (L1, PDF p. 681)

**91.** **c** The gyro is remote from the indicator and sits in the electric compartment. (L1, PDF p. 683)

**92.** **c** Two independent gyros are needed, one for each pilot's indication. (L1, PDF p. 685)

**93.** **b** The attitude data is also provided to the autoflight system and to the weather radar system. (L1, PDF p. 685)

**94.** **b** The two main values are the attitude, shown as an artificial horizon, and the flight director indications. (L1, PDF p. 687)

**95.** **b** The attitude director indicator is part of the Basic T. (L1, PDF p. 687)

**96.** **a** The slip indicator is purely mechanical: a ball in a liquid filled curved glass tube. (L1, PDF p. 689)

**97.** **c** The indicator represents a rate of turn movement of 180 degrees per minute. (L1, PDF p. 689)

**98.** **c** It is there as an additional safety feature. (L1, PDF p. 693)

**99.** **b** Fast erection is run only when the gyro has been without current for more than 30 seconds. (L2, PDF p. 681)

**100.** **a** Those are exactly the conditions that end fast erection and drop the motor to reduced voltage. (L2, PDF p. 681)

**101.** **a** Strong horizontal acceleration switches the roll erector off, the roll cut-off, and strong longitudinal acceleration switches the pitch erector off. (L2, PDF p. 681)

**102. b** The pitch erector cut-off lasts a maximum of three minutes. (L2, PDF p. 681)

**103. c** The flag monitor requires VG power greater than 90 V before the valid voltage is given out. (L2, PDF p. 681)

**104. b** Synchros on the gimbals convert the mechanical pitch and roll angles into electrical signals. (L2, PDF p. 683)

**105. b** The valid signal is sent only while the gyro is operating at normal speed. (L2, PDF p. 683)

**106. b** A switching brings the third vertical gyro onto the indicator that lost its own gyro. (L2, PDF p. 685)

**107. a** The third gyro can be switched to whichever of the two ADIs has lost its own source. (L2, PDF p. 685)

**108. b** Being part of the housing, the symbol moves with the airframe it is bolted to. (L2, PDF p. 687)

**109. b** The ball runs counter to the aircraft and the vertical gyro holds it level. (L2, PDF p. 687)

**110. a** The valid direct current is the gyro's statement that it is working. (L2, PDF p. 687)

**111. b** Gravity alone settles the ball, which is why it shows the apparent vertical. (L2, PDF p. 689)

**112. a** Its movement answers changes of rotational velocity about the vertical axis. (L2, PDF p. 689)

**113. a** It is a moving coil instrument. (L2, PDF p. 689)

**114. b** The localizer marks are about one degree of deviation each. (L2, PDF p. 689)

**115. a** The bars command the direction to fly, not the direction being flown. (L2, PDF p. 689)

**116. b** The signal is derived directly from the indication rather than from the gyro feeding it. (L2, PDF p. 693)

**117. c** A difference of several degrees is what triggers the alert. (L2, PDF p. 693)

**118. b** The instrument warning system is the one that triggers the alert. (L2, PDF p. 693)

**119. b** The message is CHECK ATTITUDE and it appears on top of the attitude indication on both displays. (L2, PDF p. 693)

**120. a** The third reference is either vertical gyro number 3 or the standby horizon. (L2, PDF p. 693)

**121. a** Reporting valid marks the end of fast erection, and from then on the erectors run at the reduced rate of 2°/min. (L3, PDF p. 681)

**122. b** Of the four flag monitor conditions only completion of fast erection is still outstanding in this case. (L3, PDF p. 681)

**123. a** A duplicated system such as a second autopilot takes its attitude from the first officer's gyro, the first system taking the captain's. (L3, PDF p. 685)

**124. c** Pitch is answered by the ball itself rotating about its own axis inside the fork. (L3, PDF p. 687)

**125. c** The attitude half states the present attitude and the flight director half commands the attitude to fly, so together they give present state and demand. (L3, PDF p. 689)

**126. a** The symbol doubles as the radio altimeter indication, and its travel up towards the aeroplane means the radio altitude is falling. (L3, PDF p. 689)

**127. b** The turn indication is driven by yaw rate and is always paired with the ball's slip indication. (L3, PDF p. 689)

**128. b** The flag is expected during the test because null monitoring brings it in. (L3, PDF p. 691)

**129. a** That deliberate tilt is the response to a lost power supply or a missing valid signal. (L3, PDF p. 691)

**130. c** All four listed conditions must hold together, and the power supply is one of them. (L3, PDF p. 691)

**131. a** Two disagreeing sources can never settle themselves; a third independent source is always needed. (L3, PDF p. 693)

# Turn and slip indicators

PDF pages 695–709 · 35 questions

## LEVEL 1 — RECALL

**132.** Which turn must the aircraft be able to make in one minute when flying a holding pattern ●

a. a 180° turn
b. a 360° turn
c. a 90° turn

**133.** Which sensor is used to measure how fast the aircraft is turning ●

a. an inclinometer
b. a vertical gyro
c. a rate gyro

**134.** What damps the movement of the ball in the slip indicator ●

a. a magnetic brake
b. a light return spring
c. a liquid

**135.** Of what material is the ball of the slip indicator made ●

a. glass
b. plastic
c. metal

**136.** Which turn gives a full deflection of the rate of turn indication ●

a. a complete turn that would take two minutes
b. a complete turn that would take one minute
c. a half turn that would take one minute

**137.** What is a complete turn flown in two minutes called ●

a. a coordinated turn
b. a standard turn
c. a holding turn

**138.** Which combination of degrees of freedom does the rate of turn indicator have ●

a. one complete and one constrained
b. one complete and two constrained
c. two complete and none constrained

**139.** To which aircraft axis is the gyroscope axis of the rate of turn indicator parallel ●

a. the lateral axis
b. the longitudinal axis
c. the vertical axis

**140.** How many degrees of freedom does a rate gyro have ●

a. three
b. two
c. one

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**141.** Which three indicators can be used to fly the required rate of turn ●

a. the turn and slip indicator, the turn coordinator and the standby horizon
b. the turn and slip indicator, the vertical gyro and the ADI
c. the turn and slip indicator, the turn coordinator and the ADI

**142.** An aircraft is fitted with a glass cockpit. How is the rate of turn presented to its crew? ●

a. by the ball of the inclinometer
b. on an electronic display
c. on a pointer over a mechanical scale

**143.** Which instrument gives the slip indication that always goes with the rate of turn indication ●

a. an inclinometer, also called the ball
b. a rate gyro, also called the turn needle
c. a vertical gyro, also called the horizon

**144.** What does the slip indication show the pilot during a turn

a. the direction of the rate of turn
b. the direction of the resultant forces
c. the direction of the bank

**145.** What do the rate of turn pointer and the ball tell the pilot when they are read together

a. the radius of the turn
b. the bank angle of the turn
c. the quality of the turn

**146.** The ball of a slip indicator sits exactly in the centre during a turn. Why does that mean the turn is perfectly coordinated? ●

a. the rate of turn has reached its full deflection
b. the vertical axis is parallel to the resultant force
c. the vertical axis is parallel to the true vertical

**147.** The gyro of a turn indicator precesses against its return spring. On what does that precession force depend? ●

a. on the gyro stability, which follows the gyro speed
b. on the stiffness of the return spring itself
c. on the gyro stability, which follows the gyro mass

**148.** About which aircraft axis can the frame carrying the gyro axis rotate

a. about the lateral axis
b. about the longitudinal axis
c. about the vertical axis

**149.** By how much is the rate gyro of a turn coordinator turned from its normal position ●

a. by 20 degrees
b. by 90 degrees
c. by 45 degrees

**150.** Why is the rate gyro of a turn coordinator canted ●

a. so that it is also sensitive to the roll rate
b. so that it is also sensitive to the pitch rate
c. so that it is no longer sensitive to the yaw rate

**151.** For what is the rate gyro mainly used ●

a. as a highly sensitive rotary motion sensor
b. as a damper for the slip indicator ball
c. as a stable long term attitude reference

**152.** A force acts on the constrained input axis of a rate gyro. How does the gyro respond to it? ●

a. it precesses around the precession axis
b. it precesses around the input axis
c. it tilts bodily along the spin axis

**153.** By which means can the second degree of freedom of a rate gyro be limited ●

a. by a spring or by a mechanical stop
b. by a counterweight or by magnetic braking
c. by a spring or by movement dependent damping

**154.** Which input to output behaviours can a rate gyro be built to have ●

a. proportional, integral or differential
b. integral, differential or exponential
c. proportional, linear or logarithmic

**155.** What is often used in place of the spring in a rate of turn gyro ●

a. a torsion bar
b. a damping piston
c. a torque motor

**156.** Around how many of the aircraft axes can a rate of turn gyro measure rotary velocity ●

a. around one only, set by the way it is fitted
b. around two, set by its two degrees of freedom
c. around all three, whatever its orientation

**157.** How are angular velocity gyroscopes with integrated features constrained ●

a. by a spring rather than by damping
b. by viscose damping rather than a spring
c. by a torsion bar rather than a spring

**158.** Which gyro measures angles in the aircraft axes with very high precision ●

a. an attitude gyro with three degrees of freedom
b. an angular velocity gyroscope with integrated features
c. a directional gyroscope with automatic drift compensation

**159.** Towards what are loose objects inside the aircraft drawn when the aircraft slips ●

a. towards the centre of the curve
b. towards the outside of the curve
c. towards the nose of the aircraft

## LEVEL 3 — APPLY AND REASON

**160.** In a turn the ball has moved out towards the inside of the turn. What does that say about the aircraft? ●

a. the bank angle suits the rate of turn but the speed is too high
b. the bank angle is too small for the actual rate of turn
c. the bank angle is too large for the actual rate of turn

**161.** The ball has moved out in the direction of the turn. What must the pilot do to bring the turn back into balance? ●

- **a.** lower the bank angle and leave the rudder central
- **b.** lower the rate of turn with rudder towards the ball
- **c.** raise the rate of turn with rudder towards the ball

**162.** The ball has moved out opposite to the direction of the turn. What is wrong and what has to be done? ●

- **a.** the bank angle is too small, so the rate of turn must be raised
- **b.** the bank angle is too large, so the rate of turn must be raised
- **c.** the bank angle is too small, so the rate of turn must be lowered

**163.** The aircraft starts to turn about its vertical axis. What then happens inside the rate of turn indicator?

- **a.** the frame is held still and the gyro axis tilts the pointer directly
- **b.** the gyro axis precesses and twists the frame against the return spring
- **c.** the gyro axis stays put while the frame is turned by the return spring

**164.** An aircraft is holding a steady bank angle through a turn. What is the turn coordinator showing at that moment? ●

- **a.** the bank angle itself
- **b.** the roll rate
- **c.** the yaw rate

**165.** A pilot is rolling into a turn and the bank angle is still changing. What is the turn coordinator showing while that goes on? ●

- **a.** the yaw rate
- **b.** the resultant force of the turn
- **c.** the roll rate

**166.** A rate of turn gyro is held by a linear spring. What can then be said about its precession angle? ●

- **a.** it falls as the angular velocity about the measuring axis rises
- **b.** it is independent of the angular velocity about the measuring axis
- **c.** it is proportional to the angular velocity about the measuring axis

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## Answers — Turn and slip indicators

**132. a** The holding pattern turn is defined as 180° in one minute. (L1, PDF p. 695)

**133. c** The rate of turn sensor is a rate gyro. (L1, PDF p. 695)

**134. c** The ball's movement is damped by the liquid it sits in. (L1, PDF p. 697)

**135. c** The ball is a metal one. (L1, PDF p. 697)

**136. a** Full deflection corresponds to a complete turn in two minutes. (L1, PDF p. 699)

**137. b** A turn in which a complete turn would need two minutes is called a standard turn. (L1, PDF p. 699)

**138. a** It has one complete degree of freedom and one that is constrained. (L1, PDF p. 701)

**139. a** The gyro axis lies parallel to the lateral axis. (L1, PDF p. 701)

**140. b** A rate gyro has only two degrees of freedom. (L1, PDF p. 705)

**141. c** The three named are the turn and slip indicator, the turn coordinator and the ADI. (L2, PDF p. 695)

**142. b** In a glass cockpit the rate of turn appears on a display. (L2, PDF p. 695)

**143. a** The slip indication comes from an inclinometer, known as the ball. (L2, PDF p. 695)

**144. b** It shows which way the resultant forces lie during the turn. (L2, PDF p. 695)

**145. c** Read together they state how good the turn is. (L2, PDF p. 697)

**146. b** A centred ball means the aircraft's vertical axis is parallel to the resultant force of the turn. (L2, PDF p. 697)

**147. a** The precession force depends on gyro stability, and that stability is proportional to the speed. (L2, PDF p. 699)

**148. b** The frame's limited rotation is about the longitudinal axis. (L2, PDF p. 701)

**149. c** The gyro is canted by 45 degrees. (L2, PDF p. 703)

**150. a** Canting the gyro adds sensitivity to roll rate. (L2, PDF p. 703)

**151. a** Its main use is as a very sensitive sensor of rotary motion. (L2, PDF p. 705)

**152. a** A force on the input axis is answered by precession about the precession axis. (L2, PDF p. 705)

**153. c** The limitation can be achieved with a spring or with movement-dependent damping. (L2, PDF p. 705)

**154. a** The three behaviours named are proportional, integral and differential. (L2, PDF p. 705)

**155. a** A torsion bar often replaces the spring. (L2, PDF p. 705)

**156. c** All three aircraft axes can be measured, and the gyro's orientation does not matter. (L2, PDF p. 705)

**157. b** They use viscose damping in place of a spring. (L2, PDF p. 705)

**158. b** Angular velocity gyroscopes measure angles of attitude in the aircraft axes with very high precision, in contrast to attitude gyros. (L2, PDF p. 705)

**159. a** In a slip the devices inside are drawn in towards the centre of the curve. (L2, PDF p. 709)

**160. c** A ball out in the direction of the turn means too much bank for the rate of turn being flown. (L3, PDF p. 697)

**161. c** With the ball out into the turn the turn rate must be raised, using rudder towards the ball. (L3, PDF p. 697)

**162. c** A ball out the other way means too little bank for the rate flown, and the cure is to lower the rate of turn. (L3, PDF p. 697)

**163. b** The turn about the vertical axis makes the gyro precede, and that precession twists the frame against the spring. (L3, PDF p. 701)

**164. c** With the bank angle constant the coordinator reads yaw rate. (L3, PDF p. 703)

**165. c** While the bank angle is changing the coordinator reads roll rate instead. (L3, PDF p. 703)

**166. c** A linear spring makes the precession angle proportional to the angular velocity about the measuring axis. (L3, PDF p. 705)

# Directional gyros

PDF pages 711–719 · 28 questions

## LEVEL 1 — RECALL

**167.** What other name is given to the direction indicator ●

a. Position indicator
b. Heading indicator
c. Attitude indicator

**168.** How many degrees of freedom does a direction indicator have, and how does its axis lie ●

a. two degrees of freedom and a horizontal axis
b. three degrees of freedom and a vertical axis
c. three degrees of freedom and a horizontal axis

**169.** Which three effects does the indication precision of a gyro compass depend on ●

a. Apparent drift, real drift and cardan error
b. Apparent drift, inclination and cardan error
c. Apparent drift, real drift and parallax error

**170.** A directional gyro has just been synchronised. For how long does its indication stay relatively exact? ●

a. 5 - 10 minutes
b. 5 - 10 seconds
c. 15 - 20 minutes

**171.** In a remote compass system, what are directional gyros used as ●

a. Amplifiers
b. Indicators
c. Sensors

**172.** Which two parts carry out the erection task in an electrical directional gyro ●

a. A level switch and a flux valve
b. A level switch and a torque motor
c. A pressure switch and a torque motor

**173.** Which task is the liquid level switch with its torque motor fitted for ●

a. Drift elimination
b. Levelling
c. Monitoring

**174.** A directional gyro component list includes a slaving torque motor. What is that motor fitted for? ●

a. To monitor the rpm
b. To level the frame
c. To eliminate drift

**175.** Which two kinds of drift does the slaving torque motor eliminate ●

a. Real drift and pole drift
b. Apparent drift and real drift
c. Apparent drift and cardan error

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**176.** The aircraft turns its nose while the direction indicator is running. What does the gyro do ●

a. it keeps its direction stable
b. it turns round with the aircraft nose
c. it is driven faster during the turn

**177.** A direction indicator has just reached its operational speed. What has to happen before its heading can be used? ●

a. The heading must be set automatically
b. The heading must be set manually
c. The heading must be set from the lubber line

**178.** What property of a space stable gyro allows it to hold the course

a. It returns to the horizontal after every turn
b. It aligns itself with the earth's magnetic lines
c. It maintains the position it has taken up

**179.** What must the rotational axis of a gyro compass remain parallel to for its directional information to be read ●

a. The aircraft's longitudinal axis
b. The earth's magnetic axis
c. The earth's surface

**180.** How is a gyro compass set, and from which indication ●

a. Automatically, from the magnetic compass
b. Manually, from the magnetic compass
c. Manually, from the flux valve

**181.** Where are the sensor and the indicator of a remote compass system installed ●

a. The sensor in the cockpit and the indicator anywhere
b. Both the sensor and the indicator in the cockpit
c. The sensor anywhere and the indicator in the cockpit

**182.** An erection system is fitted to a directional gyro. In which plane does it hold the gyro axis?

a. Vertical
b. Along the lubber line
c. Horizontal

**183.** What is the important difference between a directional gyro and a vertical gyro ●

a. The directional gyro has no erection system
b. The directional gyro is subject to drift
c. The directional gyro is never electrical

**184.** A commercial aircraft is fitted with electrical directional gyros. How is their drift dealt with? ●

a. It is compensated automatically
b. It is compensated by the vertical gyro
c. It is compensated by the pilot each hour

**185.** Why is the directional gyro more sophisticated than the gyro compass ●

a. It is part of the instrument panel, not of a remote system
b. It is part of the remote compass system, not of the panel
c. It is part of the standby compass, not of the panel

**186.** The inner frame of a directional gyro leaves the horizontal. What does the liquid level switch signal? ●

a. The heading synchro
b. The torque motor
c. The slaving amplifier

**187.** To which axis does the levelling torque motor give momentum ●

a. The axis of the gyro rotor
b. The axis of the inner frame
c. The axis of the outer frame

**188.** What produces the signals received by the slaving torque motor ●

a. The earth magnetic field detector
b. The liquid level switch circuit
c. The heading synchro amplifier

**189.** A directional gyro is running up after power-on. What rotor speed must be reached before a valid signal is passed?

a. 75%
b. 100%
c. 50%

## LEVEL 3 — APPLY AND REASON

**190.** A pilot is about to adjust the direction indicator after start-up. Which instrument has to be read first? ●

a. The magnetic compass
b. The turn indicator
c. The attitude indicator

**191.** A crew finds the gyro compass reading has drifted away from the magnetic compass reading after some minutes of flight. What does that call for? ●

a. Re-caging the outer frame by hand at regular intervals
b. Re-levelling the inner frame by hand at regular intervals
c. Synchronising the course indication by hand at intervals

**192.** A directional gyro in a remote compass system is found to have no slaving torque motor fitted. What does this mean for its drift?

a. It is removed by the levelling torque motor
b. It cannot be compensated at all
c. It is compensated by other means

**193.** A directional gyro has begun to drift and the slaving momentum is applied to the inner frame axis. Until what point does the slaving continue? ●

a. Until the inner frame is back in the horizontal
b. Until the gyro heading matches the flux valve heading
c. Until the gyro heading matches the standby compass heading

**194.** The servo system of the indicator does not compare correctly. What does the monitoring system of the directional gyro do?

a. It passes no valid signal to the consumer
b. It passes the flux valve heading instead
c. It passes a valid signal to the consumer anyway

## Answers — Directional gyros

**167. b** The direction indicator is also called the heading indicator. (L1, PDF p. 711)

**168. c** The direction indicator has three degrees of freedom together with a horizontal axis. (L1, PDF p. 711)

**169. a** Indication precision is limited by apparent drift, real drift and cardan error. (L1, PDF p. 713)

**170. a** The indication of the directional gyro is relatively exact only for 5 - 10 minutes. (L1, PDF p. 713)

**171. c** In a remote compass system the directional gyro is the sensing element. (L1, PDF p. 715)

**172. b** The electrical gyro erects with a level switch driving a torque motor. (L1, PDF p. 715)

**173. b** The level switch and its torque motor are the levelling pair. (L1, PDF p. 717)

**174. c** The slaving torque motor is listed against drift elimination. (L1, PDF p. 717)

**175. b** The slaving torque motor is fitted against apparent and real drift. (L1, PDF p. 719)

**176. a** The gyro keeps its direction stable while the aircraft turns its nose, so the scale connected to the gimbal shows the new heading at the lubber line. (L2, PDF p. 711)

**177. b** Once the gyro is up to speed the heading has to be adjusted by hand. (L2, PDF p. 711)

**178. c** Holding the position it was set to is what keeps the course indication. (L2, PDF p. 713)

**179. c** The indication can be read from the lubber line while the rotational axis remains parallel to the earth's surface. (L2, PDF p. 713)

**180. b** The gyro compass is set by hand to match the magnetic compass. (L2, PDF p. 713)

**181. c** Remote means the sensing unit can sit anywhere while only the display is in the cockpit. (L2, PDF p. 715)

**182. c** For a directional gyro the erection system holds the axis horizontal. (L2, PDF p. 715)

**183. b** Drift is the behaviour that sets the directional gyro apart from the vertical gyro. (L2, PDF p. 715)

**184. a** Electrical gyros in commercial aircraft carry automatic drift compensation. (L2, PDF p. 715)

**185. b** Being a remote system component rather than a panel instrument is the reason given for its greater complexity. (L2, PDF p. 717)

**186. b** The level switch output goes to the torque motor. (L2, PDF p. 719)

**187. c** Levelling momentum is applied to the outer frame axis until the inner frame is horizontal. (L2, PDF p. 719)

**188. a** The slaving signals come from an earth magnetic field detector. (L2, PDF p. 719)

**189. a** The monitoring condition for rotor speed is 75%. (L2, PDF p. 719)

**190. a** The setting value comes from the magnetic compass, which the direction indicator cannot supply itself. (L3, PDF p. 711)

**191. c** Because apparent drift, real drift and cardan error spoil the precision, the course indication has to be resynchronised by hand against the magnetic compass. (L3, PDF p. 713)

**192. c** In remote compass systems the drift can also be compensated by other means than a slaving torque motor. (L3, PDF p. 717)

**193. b** Slaving runs until the gyro agrees with the flux valve, which is the magnetic reference of the system. (L3, PDF p. 719)

**194. a** A correct servo comparison is one of the conditions for a valid signal, so failing it blocks the signal. (L3, PDF p. 719)

# Terrestrial magnetism and the direct reading compass

PDF pages 723–739 · 38 questions

## LEVEL 1 — RECALL

**195.** In which direction do compass systems always point ●

a. True north
b. Magnetic north
c. Compass north

**196.** What is the angle between true north and magnetic north called ●

a. Inclination
b. Deviation
c. Variation

**197.** What represents the aircraft longitudinal axis in a heading indication ●

a. The magnetic needle
b. The lubber line
c. The compass scale

**198.** Which magnetic field lines can a compass use to measure the heading ●

a. The horizontal ones
b. The declining ones
c. The vertical ones

**199.** Where are all the magnetic field lines horizontal ●

a. Near the isogonic lines
b. Near the equator
c. Near the poles

**200.** Which three methods are used to measure the angle between the aircraft longitudinal axis and magnetic north ●

a. Mechanical, electronic and mathematical
b. Electronic, mathematical and gyroscopic
c. Mechanical, electrical and magnetic

**201.** Which electrical part does a direct reading compass contain ●

a. A torque motor
b. A light
c. A detector coil

**202.** Where is the standby compass usually located ●

a. On the centre panel below the front windshields
b. On the overhead panel above the front windshields
c. At the centerpost between the front windshields

**203.** On which law of magnetism is the function of the magnetic compass based ●

a. Unlike poles attract and like poles repel
b. Like poles attract and unlike poles repel
c. All poles attract when they are close enough

**204.** What kind of fluid is used in the compass housing to inhibit the movement of the card ●

a. A water or glycol based fluid
b. A silicone or ester based fluid
c. A mineral or alcohol based fluid

**205.** To what does the aircraft magnetic field change the direction of magnetic north ●

a. Grid north
b. True north
c. Compass north

**206.** What is the difference between magnetic north and compass north called ●

a. Inclination
b. Variation
c. Deviation

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**207.** A magnet has magnetic lines running between its poles. What is the effective area of those lines called? ●

a. The magnetic axis
b. The magnetic pole
c. The magnetic field

**208.** How may the earth be regarded in terms of its magnetism ●

a. As a giant permanent magnet with a bar magnet interior
b. As a giant electromagnet with a wound coil interior
c. As a giant permanent magnet with a hollow interior

**209.** What does a suspended, free-turning magnetic needle align itself with ●

a. The earth's magnetic lines
b. The aircraft's longitudinal axis
c. The earth's magnetic axis

**210.** An aircraft is flown to a different part of the world. What does the actual value of the variation depend on?

a. The heading being flown
b. The aircraft's own magnetic field
c. The position on earth

**211.** Between which two directions is the indicated heading measured ●

a. The aircraft longitudinal axis and the direction to the north pole
b. The lubber line and the aircraft longitudinal axis
c. The aircraft lateral axis and the direction to the north pole

**212.** Why do the different variations at different locations cause no problem in practice

a. Navigation equipment removes the local variation automatically
b. Navigation equipment, runways and maps all use the magnetic heading
c. Navigation equipment, runways and maps all use the true heading

**213.** Why must runway headings be updated every few years ●

a. The position of the magnetic poles is drifting
b. The position of the geographic poles is drifting
c. The variation is zero along the isogonic lines

**214.** The earth's magnetic field is being resolved at a given latitude. Into which two components is it split?

a. A horizontal and a vertical component
b. A magnetic and a geographic component
c. A horizontal and a lateral component

**215.** An aircraft measures its heading by the electronic method. What determines the angle? ●

a. A stored variation value
b. A freely rotating permanent magnet
c. An earth magnetic field detector

**216.** Why is the direct reading compass called the standby compass in commercial aircraft ●

a. Its indication belongs to the basic T and must always be available
b. Its indication replaces the gyro indication during every turn
c. Its indication belongs to the basic T and is the primary one in flight

**217.** Why is the standby compass placed at the centerpost ●

a. Only one pilot can read it but foreign magnetic fields are distant
b. Both pilots can read it and foreign magnetic fields are distant
c. Both pilots can reach it and direct sunlight is avoided

**218.** The scale of a standby compass takes up the direction of the field. What turns the scale? ●

a. A permanent magnet
b. A torque motor
c. An electromagnet

**219.** Why is the direct reading compass filled with a fluid ●

a. To lubricate the compass scale bearings
b. To insulate the compass scale from heat
c. To dampen the compass scale movements

**220.** With what does the freely moving magnet of a compass line up ●

a. The longitudinal axis of the aircraft
b. The vertical components of the earth's field
c. The horizontal components of the earth's field

**221.** What does the compass card under the lubber line show ●

a. The magnetic field against the aircraft's longitudinal axis
b. The longitudinal axis against the earth's magnetic pole
c. The longitudinal axis against the earth's geographic pole

**222.** What condition must be met for the lubber line to represent the aircraft longitudinal axis ●

a. The compass must be correctly compensated
b. The compass must be correctly filled
c. The compass must be correctly installed

**223.** The earth's magnetic field is weak. What does this mean for the compass magnet?

a. A high inclination of the compass magnet
b. A high directive force of the compass magnet
c. A low directive force of the compass magnet

**224.** The damping medium changes volume with temperature. What balances this out? ●

a. Washers
b. Slots
c. Bellows

**225.** What inside the aircraft also affects the direct reading compass ●

a. A magnetic field from electrical equipment
b. A vibration field from electrical equipment
c. A magnetic field from the earth's interior

**226.** A standby compass is found to have a compass error from the airframe. What is used to minimize it? ●

a. Small adjustable magnets inside the compass
b. Copper washers under the mounting plate
c. Small adjustable weights inside the compass

### LEVEL 3 — APPLY AND REASON

**227.** Why does the north pole of the magnetic needle show north ●

a. The magnetic south pole of the earth is there
b. The magnetic north pole of the earth is there
c. The geographic south pole of the earth is there

**228.** An aircraft carries a modern inertial reference unit. Why is a compass reference carried as well? ●

a. Radio indications are referenced to magnetic north
b. Inertial units cannot produce a heading at all
c. Inertial units are accurate for a short time only

**229.** An aircraft is to operate in a region close to the magnetic pole. Which heading source must be used there and why? ●

a. A flux valve, because the vertical component takes over
b. A directional gyro, because the horizontal component is too weak
c. A standby compass, because the horizontal component is strongest

**230.** A compass finds the angle with a mounted, freely rotating permanent magnet. Which method is this and which kind of compass? ●

a. The electronic method, a remote reading compass
b. The mechanical method, a remote reading compass
c. The mechanical method, a direct reading compass

**231.** An aircraft finds its heading from data held in a computer rather than by sensing the field. Which method is this and what does it use? ●

a. The mathematical method, using the stored variation for each location
b. The electronic method, using the field detector fitted at each location
c. The mechanical method, using the magnet mounted at each location

**232.** Fluid damping has been accepted so the indicator stays stable under vibration. What has to be accepted along with it?

a. A loss of directive force of the magnet
b. A time delay after a change of direction
c. An increase of the installation error

## Answers — Terrestrial magnetism and the direct reading compass

**195. b** Compass systems follow the earth's magnetic field, which defines magnetic north. (L1, PDF p. 723)

**196. c** Variation is the name for the angle between true north and magnetic north. (L1, PDF p. 725)

**197. b** The lubber line stands for the longitudinal axis of the aircraft. (L1, PDF p. 725)

**198. a** Only the horizontal field lines are usable for heading measurement. (L1, PDF p. 731)

**199. b** The field lines run horizontal only in the equatorial region. (L1, PDF p. 731)

**200. a** The three methods of measuring the angle between the aircraft's longitudinal axis and magnetic north are mechanical, electronic and mathematical. (L1, PDF p. 733)

**201. b** The only electrical item in a direct reading compass is its light. (L1, PDF p. 735)

**202. c** The usual position is the centerpost between the two front windshields. (L1, PDF p. 735)

**203. a** The main law of magnetism quoted is that unlike poles attract and like poles repel. (L1, PDF p. 737)

**204. c** The damping medium named is a mineral or alcohol based fluid. (L1, PDF p. 737)

**205. c** The shifted direction is named compass north. (L1, PDF p. 739)

**206. c** Deviation is the name for that difference. (L1, PDF p. 739)

**207. c** The area over which the lines act is the magnetic field. (L2, PDF p. 723)

**208. a** The earth can be regarded as a giant permanent magnet whose interior is a bar magnet with its ends at the magnetic poles. (L2, PDF p. 723)

**209. a** A free needle takes up the direction of the earth's magnetic lines, which is what makes it usable for navigation. (L2, PDF p. 723)

**210. c** Variation is a property of where the aircraft is, not of how it is flown. (L2, PDF p. 725)

**211. a** Heading is the angle from the aircraft longitudinal axis to north. (L2, PDF p. 725)

**212. b** Everything is referenced the same way, to magnetic heading, so the local value does not have to be applied. (L2, PDF p. 727)

**213. a** Because the magnetic poles move, anything referenced to them goes out of date. (L2, PDF p. 727)

**214. a** The field is resolved into horizontal and vertical components. (L2, PDF p. 731)

**215. c** The electronic method senses the field with an earth magnetic field detector. (L2, PDF p. 733)

**216. a** Being part of the basic T, the heading has to remain available at all times, which is the standby role. (L2, PDF p. 735)

**217. b** The position suits both crew members and keeps the compass clear of equipment fields. (L2, PDF p. 735)

**218. a** The scale is driven by a permanent magnet aligning with the field. (L2, PDF p. 735)

**219. c** The fluid is there as a damping medium for the scale. (L2, PDF p. 735)

**220. c** The compass magnet follows the horizontal components of the earth's field. (L2, PDF p. 737)

**221. b** The reading at the lubber line is the longitudinal axis direction against the earth's magnetic pole. (L2, PDF p. 737)

**222. c** The lubber line only stands for the longitudinal axis if the installation is correct. (L2, PDF p. 737)

**223. c** A weak field gives the compass magnet only a weak directive force. (L2, PDF p. 737)

**224. c** Bellows take up the expansion and contraction of the damping fluid. (L2, PDF p. 737)

**225. a** Equipment running in normal operation sets up a magnetic field inside the aircraft. (L2, PDF p. 739)

**226. a** The compass compensation consists of small adjustable magnets inside the instrument. (L2, PDF p. 739)

**227. a** What lies at the earth's north is a magnetic south pole, and unlike poles attract. (L3, PDF p. 723)

**228. a** Even with INS or IRS fitted, indications such as the VOR RMI are referenced to magnetic north. (L3, PDF p. 723)

**229. b** Inclination leaves too little horizontal field at high latitude, so a gyro or inertial reference replaces the compass. (L3, PDF p. 731)

**230. c** The magnetic needle principle is the mechanical method and gives the direct reading compass. (L3, PDF p. 733)

**231. a** Working from stored variation values in a navigation computer is the mathematical method. (L3, PDF p. 733)

**232. b** The time delay following a change of direction is an unavoidable disadvantage of keeping the indicator stable against vibration. (L3, PDF p. 737)

# Compass compensation and swinging

PDF pages 741–755 · 25 questions

## LEVEL 1 — RECALL

**233.** A compass is about to be compensated. Into which classes are its errors divided? ●

a. A, B and D errors
b. B, C and D errors
c. A, B and C errors

**234.** A compass is found to carry a B error — in which directions does it show up? ●

a. in all directions
b. in the east-west direction
c. in the north-south direction

**235.** An engineer is picking out tools for a compass compensation — which ones may be used? ●

a. only approved antimagnetic tools
b. only calibrated torque tools
c. only insulated electrical tools

**236.** Within what tolerance must the aircraft be aligned to North - South before the SCC is used ●

a. +/− 2 degrees
b. +/− 10 degrees
c. +/− 5 degrees

**237.** What is required of the deviation that is left once a compensation has been performed ●

a. it should be as small as possible
b. it should be below one degree on every heading
c. it should be exactly zero on every heading

**238.** In what steps are the headings checked when a deviation card is made up ●

a. in steps of 10 degrees
b. in steps of 45 degrees
c. in steps of 30 degrees

**239.** An aircraft is flying over the northern part of the earth — in which direction does the vertical component of the field point there? ●

a. downwards
b. along the meridian
c. upwards

**240.** In which direction of flight does an acceleration or a deceleration leave the heading indication unchanged ●

a. a northerly direction
b. a westerly direction
c. an easterly direction

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**241.** Why does a magnetic compass or a remote compass system have to be compensated at all

a. the earth's field loses strength as the aircraft climbs away from the ground
b. the vertical field component pulls the magnet out of the horizontal plane
c. aircraft ferro- and electro-magnetism sets up fields that cause a deviation

**242.** Some error is still left after a compass has been compensated — where is that remainder written down? ●

a. on the compensating disc at the airport site
b. on a deviation card carried on the aircraft
c. on a label fitted beside the flux valve

**243.** An indication is falsified by the amount of the deviation — what is that heading called? ●

a. the compass heading
b. the true heading
c. the magnetic heading

**244.** An engineer walks out to compensate a compass. Which of the items he is carrying can deflect the compass?

a. his ear defenders, his gloves and his torch
b. his glasses, his watch and his mobile phone
c. his notebook, his pen and his pass

**245.** What can be done with the Standby Compass Calibrator without physically turning the aircraft ●

a. measure the vertical component of the earth's field at the compass
b. compensate permanent magnetisation errors and complete the deviation card
c. set the directional gyro to agree with the remote compass heading

**246.** Which two separate functions does the SCC perform

a. it cancels the aircraft's permanent field and adds a vertical field
b. it rotates the standby compass and records the resulting deviation
c. it cancels the local earth's field and adds a rotatable standard field

**247.** The aircraft cannot be put exactly on North - South for a standby compass calibration — what is done about the residual misalignment? ●

a. it is entered on the deviation card
b. it is taken out on the E–W screw
c. it is measured and set into the SCC

**248.** A compensation is finished and a small deviation is still there — why can it not be driven to zero?

a. the compensation is only checked on two headings
b. the means of adjustment cannot do any better
c. the earth's field changes while the aircraft is turned

**249.** The readings have been taken on every checked heading — what is done with each of them? ●

a. each one is recorded in the compensation log
b. each one is recorded on a deviation card
c. each one is set into the remote compensator

**250.** A compass magnet is kept level on its bearing — which component of the earth's field does it then measure?

a. the horizontal component
b. the vertical component
c. the total resultant field

**251.** What decides whether a tilted compass turns to the right or to the left ●

a. the actual heading of the aircraft
b. the actual bank angle of the aircraft
c. the actual airspeed of the aircraft

**252.** A compass has been tilted out of the horizontal plane during an acceleration — what does the crew see? ●

- **a.** a slowly spinning card
- **b.** a frozen heading
- **c.** a falsified heading

**253.** Why is there no heading error when the aircraft accelerates on a northerly heading ●

- **a.** the magnet is held exactly horizontal by its bearing on that heading
- **b.** the vertical field acts on the magnet the way the horizontal field does
- **c.** the vertical field has no component at all at that position on the earth

**254.** Why does a southerly heading behave like a northerly one during an acceleration ●

- **a.** the permanent magnet points in the same direction
- **b.** the pivot point bearing locks the compass scale
- **c.** the vertical component reverses with the hemisphere

### LEVEL 3 — APPLY AND REASON

**255.** Compensation is being worked from INS/IRS data alongside the aircraft's own heading sources — which source supplies the magnetic heading?

- **a.** the INS/IRS
- **b.** the standby compass
- **c.** the remote compass system

**256.** A remote compass is 4 ° off the target course on one of the four main headings. What does that mean for the compensation?

- **a.** it is out of tolerance and must be adjusted further
- **b.** it is in tolerance and the 30° headings can be started
- **c.** it is in tolerance for a main heading but not for a 30° heading

**257.** An acceleration has tilted a compass out of the horizontal plane — what now acts on its magnet? ●

- **a.** the permanent magnetism of the aircraft structure
- **b.** the vertical component of the earth's magnetic field
- **c.** the horizontal component of the earth's magnetic field

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## Answers — Compass compensation and swinging

**233. c** The errors to be eliminated by compensation are classed as A, B and C errors. (L1, PDF p. 741)

**234. b** A B error is the east-west error. (L1, PDF p. 741)

**235. a** Compensation may only be done with approved antimagnetic tools. (L1, PDF p. 743)

**236. a** The alignment requirement is exactly +/- 2 degrees of North - South. (L1, PDF p. 749)

**237. a** The requirement is that the deviation is as small as possible. (L1, PDF p. 751)

**238. c** Headings are checked every 30 degrees. (L1, PDF p. 751)

**239. a** Over the northern part the vertical component points down. (L1, PDF p. 753)

**240. a** On a northerly heading an acceleration or deceleration leaves the indication alone. (L1, PDF p. 755)

**241. c** Interference fields from the aircraft's own ferro- and electro-magnetism produce the deviation. (L2, PDF p. 741)

**242. b** The remaining deviation goes on the deviation card that stays on the aircraft. (L2, PDF p. 741)

**243. a** The falsified indication is named the compass heading. (L2, PDF p. 741)

**244. b** Glasses, a watch and a mobile phone are the items named as causing deflection. (L2, PDF p. 743)

**245. b** The SCC compensates permanent magnetisation errors and completes the deviation card with the aircraft standing still. (L2, PDF p. 747)

**246. c** The SCC cancels the local earth's field and adds a standard field that can be rotated. (L2, PDF p. 749)

**247. c** The misalignment is measured and fed into the SCC itself. (L2, PDF p. 749)

**248. b** Zero cannot practically be reached with the available means of adjustment. (L2, PDF p. 751)

**249. b** Each checked heading's reading goes onto the deviation card. (L2, PDF p. 751)

**250. a** Holding the magnet horizontal lets it measure the horizontal component. (L2, PDF p. 753)

**251. a** The direction of the swing depends on the heading the aircraft is on. (L2, PDF p. 753)

**252. c** The result at the instrument is a falsified heading. (L2, PDF p. 753)

**253. b** Both field components pull the magnet the same way, so nothing moves. (L2, PDF p. 755)

**254. a** On south the permanent magnet still points the same way, so the same reasoning holds. (L2, PDF p. 755)

**255. c** The remote compass gives magnetic heading, the INS/IRS gives true heading. (L3, PDF p. 743)

**256. a** The remote compass tolerance is + / − 2 °, so 4 ° is outside it and more adjustment is needed. (L3, PDF p. 745)

**257. b** Once the compass is off the horizontal the vertical component starts to influence the magnet. (L3, PDF p. 753)

# The flux valve and remote compass systems

PDF pages 757–767 · 14 questions

## LEVEL 1 — RECALL

**258.** A remote compass system is fitted to an aircraft. Which two sensing devices does it use? ●

a. a sensor for the earth's magnetic field and a vertical gyro
b. a sensor for the aircraft's own field and a rate gyro
c. a sensor for the earth's magnetic field and a directional gyro

**259.** Where is the indicator of a remote compass system fitted ●

a. in a wing tip fairing
b. in the avionics compartment
c. in the instrument panel

**260.** What does the flux valve turn the direction of the magnetic field into ●

a. a proportional mechanical rotation
b. a proportional electrical signal
c. a proportional change of resistance

## LEVEL 2 — UNDERSTAND AND EXPLAIN

**261.** A large aeroplane carries two separate remote compass systems — how are they shared out?

a. one for each engine
b. one for each pilot
c. one for each gyro platform

**262.** What does the signal detected by the flux valve represent ●

a. the strength of the earth's vertical field at the aircraft's longitudinal axis
b. the direction of the earth's horizontal field to the aircraft's longitudinal axis
c. the angle between the directional gyro and the aircraft's longitudinal axis

**263.** A flux valve has to stay level as the aircraft manoeuvres — how is it suspended in its housing? ●

a. on a full cardanic suspension
b. on a needle point bearing
c. on three rubber mounts

**264.** On what do the intensity and the direction of the sensing coil signals depend

a. on where the magnetic field points
b. on how fast the aircraft is turning
c. on how strong the excitation current is

**265.** Under which condition does a coil induce a voltage at all ●

a. only when the iron core has reached full saturation
b. only when the intensity of the magnetic field is highest
c. only when the intensity of the magnetic field changes

**266.** How is the iron core of the flux valve made to change its magnetic state ●

a. it is frequently demagnetised by a direct current in a sensing coil
b. it is frequently turned by a small motor inside the housing
c. it is frequently saturated by an alternating current in an excitation coil

**267.** What gives the flux valve its name ●

a. the alternate switching of the magnetic field
b. the liquid that flows through its hemisphere
c. the vent valve that breathes with altitude

### LEVEL 3 — APPLY AND REASON

**268.** An aircraft is held in a 40° banked turn — what happens to the flux valve measuring plane? ●

a. it stays horizontal because the cardan takes any angle
b. it is driven back to horizontal by the directional gyro
c. it can no longer stay horizontal

**269.** The excitation current has just fallen away and the core has come out of saturation — what does the magnetic field do?

a. it takes the shortest way and bypasses the three coils
b. it takes an even longer way through the iron and the coils
c. it collapses completely until the core saturates again

**270.** A flux valve output and an ordinary synchro output are compared on an oscilloscope. How does the sinusoidal voltage pattern of the flux valve differ?

a. it is shifted near to the peak value of the excitation voltage
b. it is clipped at the saturation level of the iron core
c. it is shifted near to the zero value of the excitation voltage

**271.** A compass coupler is getting no setting signal at all — which chain has broken down?

a. the directional gyro to flux valve drive path
b. the compass coupler to excitation coil feed path
c. the flux valve to control transformer path

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## Answers — The flux valve and remote compass systems

**258. c** The system pairs an earth's field sensor with a directional gyro. (L1, PDF p. 757)

**259. c** The indicator sits in the instrument panel. (L1, PDF p. 757)

**260. b** The output is an electrical signal proportional to the field direction. (L1, PDF p. 759)

**261. b** The two systems are split one per pilot. (L2, PDF p. 757)

**262. b** The flux valve signal gives the direction of the earth's horizontal field relative to the aircraft's longitudinal axis. (L2, PDF p. 757)

**263. a** It hangs in a full cardan suspension. (L2, PDF p. 757)

**264. a** Both the intensity and the direction follow the direction of the magnetic field. (L2, PDF p. 759)

**265. c** A changing field intensity is what induces the voltage. (L2, PDF p. 759)

**266. c** An alternating current in the excitation coil repeatedly saturates the core. (L2, PDF p. 759)

**267. a** The field is switched on and off like a valve, which is where the name comes from. (L2, PDF p. 759)

**268. c** The cardan only holds the plane level out to about 30°, so at 40° of bank it is exceeded. (L3, PDF p. 757)

**269. b** Unsaturated iron offers a far easier path, so the field runs the longer way through the iron and the coils. (L3, PDF p. 759)

**270. c** In the flux valve the pattern sits near the zero value of the excitation voltage. (L3, PDF p. 767)

**271. c** The path runs flux valve to control transformer to the directional gyro or compass coupler, so a loss at the coupler points back along that path. (L3, PDF p. 767)